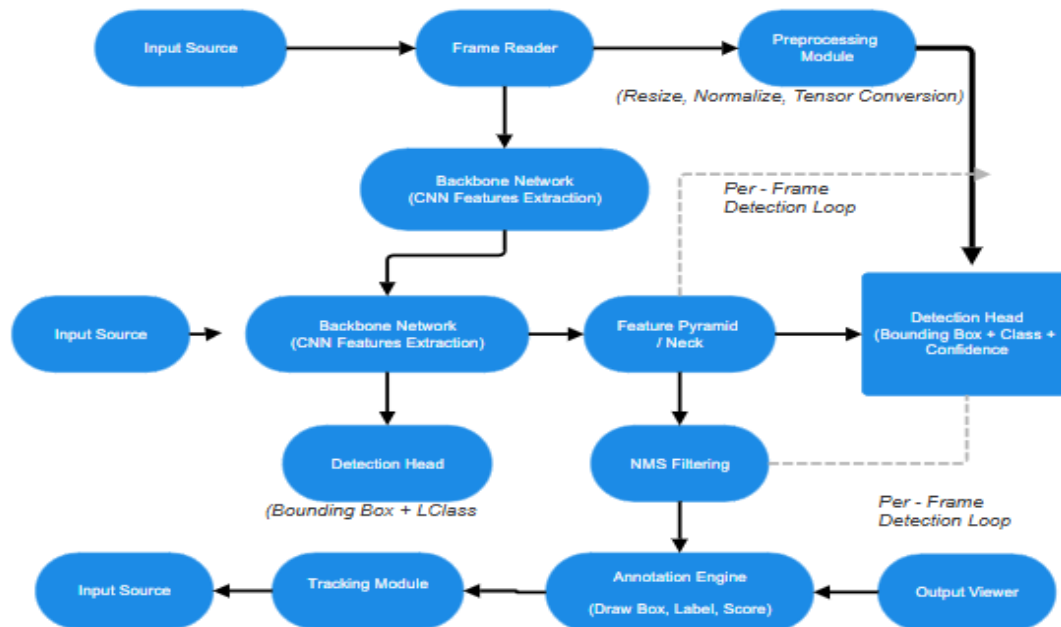


OBJECT DETECTION IN VIDEOS USING YOLO ULTRALYTICS

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1) Design and implement the pipeline for object detection in Videos using Yolov8



2) Your design should capture Object Detection in frames (yolov8) and tracking the objects across frames (tracking algorithm). Implementation - yolov8 model from Ultralytics

Code:

```
import cv2

from ultralytics import YOLO

model = YOLO("yolov8n.pt") # you can change model

video_path = "video.mp4"

cap = cv2.VideoCapture("Yolovid.mp4")

if not cap.isOpened():
    print("Error opening video")
    exit()

while True:
    success, frame = cap.read()

    if not success:
        break
```

```

results = model.track(
    frame,
    persist=True,
    tracker="botsort.yaml"
)
annotated_frame = results[0].plot()
cv2.imshow("YOLOv8 Tracking", annotated_frame)
# Press ESC to stop
if cv2.waitKey(1) & 0xFF == 27:
    break
cap.release()
cv2.destroyAllWindows()

```

Output

